

Shubh Kapoor

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TECHNICAL SKILLS

AI/ML: Python, PyTorch, scikit-learn, Deep Learning, Computer Vision, NLP, Transformers, LoRA/QLoRA

LLMs & RAG: LLM APIs, RAG, Agentic RAG, Hybrid Search, GraphRAG, Tool Calling, Structured Outputs, Embeddings, Reranking, LangGraph, LangChain, MCP, A2A

AI Evaluation: Agent Evals, Benchmarking, LLM-as-Judge, RAGAS, Regression Testing, Guardrails, Failure Analysis

MLOps & AI Systems: Docker, Kubernetes, Git/GitHub, vLLM, SGLang, LiteLLM, MLflow, Model Serving, LLM Inference, Model Routing, Quantization, OpenTelemetry, Langfuse, CI/CD

Software Engineering: C++, DSA, FastAPI, REST APIs, TypeScript, React.js, Node.js, PostgreSQL, Redis, pytest

Cloud & Infrastructure: AWS, AWS Bedrock, Distributed Systems, Concurrency, Microservices, Event-Driven Architecture, API Design, Caching, Queues, System Design

PROJECTS

Sentinel-47: AI Energy Supply Risk Intelligence Platform *Next.js, React, TypeScript, Gemini 2.5 Flash, Zustand, TanStack Query, SQLite, Tailwind CSS* [Live Demo](#)

- Engineered a weighted composite resilience index (0.30 corridor risk + 0.30 SPR + 0.25 refinery capacity + 0.15 GDP exposure) recomputed in real time across 4 coordinated risk modules.
- Implemented a SHA-256 hash-chained audit ledger with genesis-block linking and a chain-verification check, providing tamper-evident logging for every generated decision memo.
- Built a resilient data layer with 60s auto-refresh, 2x retry with backoff, and a 5-signal offline fallback dataset, isolating failures per module via error boundaries to prevent full-dashboard crashes.

AgentBreach: AI Agent Memory Security Benchmark *Python, LangChain, SQLite, Pytest, argparse*

- Built a 104-case adversarial benchmark for persistent AI-agent memory poisoning across 8 attack categories, comparing vulnerable and guarded agent configurations.
- Engineered layered defenses against prompt injection, protected-key tampering, obfuscated payloads, and malicious tool outputs, with adaptive attack testing and automated security evaluation.
- Reduced attack success rate from 26.9% to 2.9% (an 89.2% reduction) while achieving 87.5% detection recall, 100% precision, and 0% false-positive rate.

WORK EXPERIENCE

AI/ML Intern

May 2026 – Aug 2026

FintronicsAI Pvt. Ltd.

- Built a real-time driver monitoring system using YOLOv8 and MediaPipe Face Mesh; computed EAR and MAR with adaptive 3-second calibration to detect drowsiness, yawning, and distraction.
- Improved smartphone detection by curating 465 task-specific images in Roboflow from an initial 12,700-image dataset, reducing missed detections and false positives in representative driver-monitoring scenarios.
- Researched 7 financial-AI approaches spanning sentiment analysis, LSTM-DNNs, FinBERT, multimodal learning, and LLMs; proposed an adaptive multimodal framework combining news sentiment and market indicators.

UI/UX Design Intern

Dec 2025 – Jan 2026

TalentiQo Workforce Pvt. Ltd.

- Designed wireframes and high-fidelity Figma screens for 2 product features across 3 iteration cycles; created a 20-component reusable UI library covering buttons, inputs, cards, navigation, and common interaction states.
- Prepared developer handoff documentation for 2 feature flows, specifying responsive layouts, component states, spacing, typography, and key edge cases to support implementation.

EDUCATION

The LNM Institute of Information Technology

Jaipur, India

B.Tech in Computer and Communication Engineering

2023 – 2027

Dr. Virendra Swarup Education Centre

Kanpur, India

High School, Science (ISC)

2021 – 2023